FNCE 30007 DERIVATIVE SECURITIES

Tutorial 2 Questions with Solutions: Hedging with Forwards and Futures

Question 1

Does a perfect hedge always succeed in locking in the current spot price of an asset for a future transaction? Explain your answer.

Answer

No. A perfect hedge implies zero market risk. A perfect hedge does not require the hedge contract price to equal the current spot price. Consider for example the use of a forward contract to hedge a known cash inflow in a foreign currency. The forward contract locks in the forward rate not the spot rate. The spot is generally different from the forward or futures price.

Question 2

Explain why a short hedger's position improves when the basis strengthens unexpectedly and worsens when the basis weakens unexpectedly.

Answer

A short hedge is long the asset and short the futures contract. The basis is the difference between the spot price and the futures price:

$$B_t = S_t - F_t$$

The basis strengthens when the spot price increase more than or decreases less than the futures price. The value of the hedged position for the short hedger can be written as:

$$Value = F_0 + B_t$$

which increases with the basis. Intuitively, the value of the asset has increased by more than the hedge contract which benefits the short hedger.

Question 3

Imagine you are the treasurer of a Japanese company exporting electronic equipment to the U.S. Discuss how you would design a foreign exchange hedging strategy and the arguments you would use to sell the strategy to your fellow executives.

Answer

The simple answer to this question is that the treasurer should

1. *Estimate the company's future cash flows in Japanese yen and U.S. dollars*
2. *Enter into forward or futures contracts to lock in the exchange rate for the U.S. dollar cash flows.*

However, this is not the whole story. The volume of future export sales is generally not known. The company also needs to examine whether these volumes, and therefore magnitudes of the foreign cash flows, depend on the exchange rate. For example, will the company be able to raise the price of its product in U.S. dollars if the yen appreciates? If the company can do so, its foreign exchange exposure may be quite low. Or perhaps if the yen strengthens, consumer confidence falls and company exports fewer units, implying that overall profitability is even more sensitive to the exchange rate than is accounted for by the simple hedging strategy.

The key estimates are those showing the overall effect of changes in the exchange rate on the company's profitability. Once these estimates have been produced the company can choose between using forwards/futures and options to hedge its risk. The results of the analysis should be presented carefully to other executives. It should be explained that a hedge does not ensure that profits will be higher. It means that profit will be more certain. When futures/forwards are used both the downside and upside are eliminated. With options a premium is paid to eliminate only the downside.

Question 4

"For an asset where futures prices are usually less than spot prices, long hedges are likely to be particularly attractive." Explain this statement.

Answer

A company that knows it will purchase a commodity in the future is able to lock in a price close to the futures price (subject to acceptable levels of basis risk). This is likely to be attractive when the futures price is less than the spot. That said, large magnitudes of basis may reflect foregone cash flows from not holding the asset (e.g. dividends) or poor liquidity. Both would be factors that potentially mitigate the benefits of spot prices exceeding futures prices for long hedgers.

Question 5

A corn farmer argues: "I don't use futures contracts for hedging. My real risk is not the price of corn. It is my whole crop gets wiped out by the weather." Discuss this viewpoint. Should the farmer estimate his or her expected production of corn and hedge to try to lock in a price for expected production?

Answer

If weather creates a significant uncertainty about the volume of corn that will be harvested, the farmer should not enter into short forward contracts to hedge the price risk on his or her expected production. If the farmer cannot accurately predict their production, the farmer may find themselves under- or over-hedged depending on the weather situation.

In this particular case, if the weather is bad and the farmer's production is lower than expected, they are over-hedged, implying they have a net short position in the futures market. However, other farmers are likely to have been affected similarly. Corn production overall will be low and as a consequence the price of corn will be relatively high. The farmer's problems arising from the bad harvest will be made worse by losses on the short futures position.

In the opposite case of a very good harvest, the farmer will be under-hedged and prices will be lower than expected. The short futures contract at the relatively higher price provides some protection against this scenario, however, with better production, the farmer is likely to be in a better financial position than in the case of bad weather, and the former scenario should be the main factor (assuming they are risk-averse).

This problem emphasizes the importance of looking at the big picture when hedging. The farmer is correct to question whether hedging price risk while ignoring other risks is a good strategy.

Question 6

On July 1, an investor holds 50,000 shares of a certain stock. The market price is $30 per share. The investor is interested in hedging against movements in the market over the next month and decides to use the September Mini S&P 500 futures contract. The index is currently 1,500 and one contract is for delivery of $50 times the index. The beta of the stock is 1.3 (where beta is computed against the S&P 500 index value). What strategy should the investor follow to achieve the minimum variance hedge ratio?

Answer

The investor should take a short position of:

$$N^* = \beta \times \frac{50k \times 30}{1.5k \times 50} = 26\ contracts$$

Question 7

After reading the Hershey case presented in the lecture, think about and answer the following two questions.

7.1 In 2024, Hershey reported substantial gains from commodity derivatives while cocoa prices were rising sharply. Does this mean that Hershey's hedging program was designed to generate profits? Explain your answer.

Answer

Hershey may have performed differently because it had an established commodity hedging program. As cocoa prices increased significantly during 2024, gains from cocoa futures and options helped offset part of the higher raw material costs. Rather than attempting to profit from price movements, the company used derivatives to reduce the financial impact of rising cocoa prices and improve earnings stability.

7.2 Although cocoa prices had fallen substantially by 2026, why might Hershey still face significant cocoa-related costs or report losses on some commodity derivative positions?

Answer

Although cocoa prices had fallen significantly by 2026, Hershey might still experience high cocoa-related costs because it may be using inventories purchased at earlier high prices or fulfilling purchase contracts signed before prices declined. In addition, some hedge positions established when cocoa prices were higher could generate mark-to-market losses as market prices fell. Therefore, lower spot prices do not necessarily translate into immediate reductions in production costs or accounting gains.